

Adapting to Delays: On the Growth and Persistence of the Slowdown Effect

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Abstract

Delays between users' actions and system effects in interactive systems can progressively slow down user behavior. Prior work demonstrated this slowdown effect but left open how it develops over longer interaction time and whether it persists once the delay is removed. We conducted two online experiments using a point-and-click game with targets that disappeared immediately after being shot and targets that disappeared after a 0.6 s delay. In Experiment 1 ($N = 50$), we extended prior research by doubling the game rounds and found that the slowdown effect continued to increase over time. In Experiment 2 ($N = 50$), we removed the delay halfway through to test persistence. While not statistically significant, descriptive results suggest a pattern consistent with a persistent slowdown effect for several game rounds after delay removal. Together, these findings highlight the importance of considering system delays not only as an immediate influence but also as a possible long-term cognitive effect.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in interaction design**.

Keywords

System Delays, Latency, Anticipatory Action Planning, Behavioral Adaptation, Ideomotor Theory

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1 Introduction and Background

Interactive systems are ubiquitous in everyday life, from social media apps on smartphones to productivity tools and entertainment platforms. Despite continuous hardware and software improvements, delays or latencies, defined as the timespan between a user's action and the system's response, are unavoidable [21]. Such delays, even when minimal, can affect performance [5, 16], reduce accuracy in selection tasks [24], and lower the user experience [11, 19, 26, 31]. Especially in time-sensitive contexts, such as racing

and action gaming, users are susceptible to delays, as split seconds determine the difference between success and failure [4, 19, 27].

Beyond these immediate effects, prior work showed that users adapt their behavior after prolonged exposure to delays. From the perspective of ideomotor theory, actions are planned and initiated by anticipating their perceptual effects [14, 22, 29]. By repeatedly performing an action and perceiving its corresponding effect, a bi-directional action-effect association is formed [8]. This means that thinking about an outcome of an action (e.g., digitally writing a word) will automatically trigger and initiate the corresponding action (e.g., finger movements) [28]. Building on this theoretical foundation, prior research demonstrated robust compatibility effects, showing that actions are initiated more quickly when anticipated effects match previously learned associations [8, 15, 17]. Extending this line of research, several studies investigated how time aspects influence associations. Dignath et al. [6] observed that participants responded slower when action feedback was delayed, even though the delay occurred only after the action was completed. Similarly, Tanaka et al. [30] demonstrated that immediate feedback facilitated faster responses compared to delayed or absent feedback. These results indicate that temporal aspects, such as effect delays, can be integrated into action plans, thereby influencing user behavior and leading participants to slow down their initiated actions. However, the persistence of action-effect associations remains an open question. So-called event files, containing action-effect associations [12], are stored in episodic memory and share structural similarities with the learning processes of Pavlovian conditioning [10, 25]. Thus, extinction of an association does not mean removal, but the formation of a new association [7]. Consequently, following changes in system properties (e.g., removal of a delay), the influence of the previous association may persist and continue to affect user behavior while the new association is gradually learned [2, 9].

Several studies in Human-Computer Interaction (HCI) reported persistent behavioral adaptations to system delays, often as secondary findings. Delays have been shown to reduce user activity in web search and interactive visual analysis tasks, with effects that influence subsequent conditions and even persist for several days after switching from high- to low-latency systems [3, 20]. Similarly, Olguín Muñoz et al. [23] reported a so-called pacing effect in wearable assistance systems, where slowed action initiation persisted after a return to a low latency condition. However, while these studies observed persistent behavioral changes across various HCI contexts, they were not designed to test the underlying cognitive mechanisms. Building directly on the theoretical foundations of ideomotor theory, Bogon et al. [1] conducted a first HCI-specific study to demonstrate that system delays are integrated into users' action planning. In their game-like target selection task, participants clicked targets that differed in their disappearance timing:



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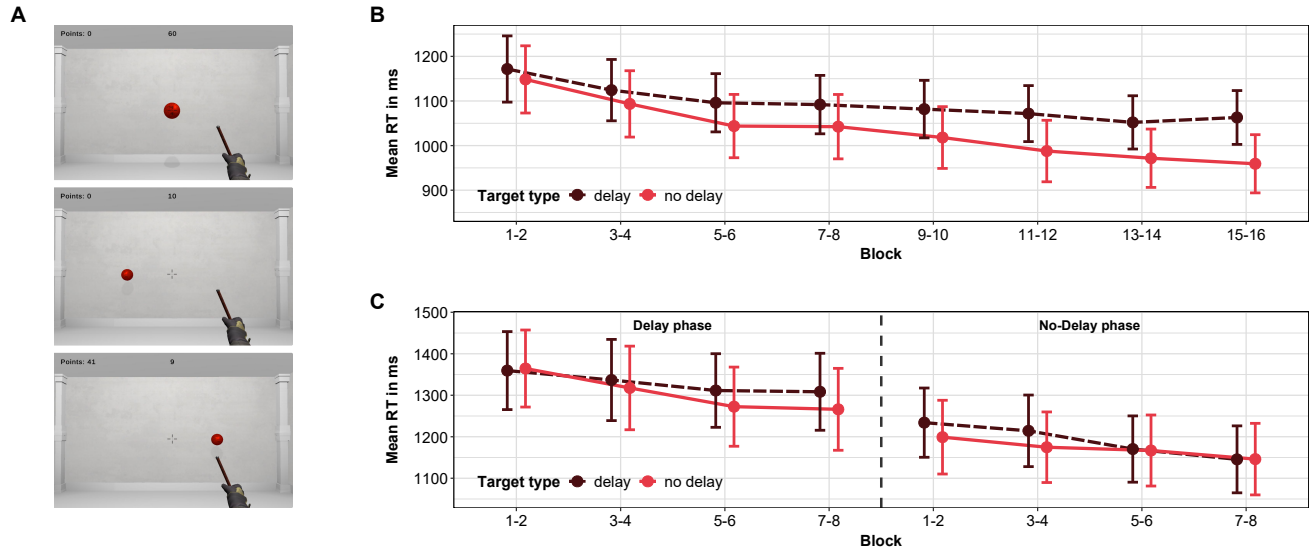


Figure 1: Game interface and response time results. (A) Game screenshots with the middle target and the side targets. (B) Mean response times (RTs) across blocks by target type in Experiment 1: Growth. Each block consists of two game rounds. (C) Mean RTs across blocks by target type and phase in Experiment 2: Persistence. Error bars indicate standard error.

targets on one side disappeared immediately upon being clicked, whereas targets on the other remained visible for 600 ms after the click. Throughout the experiment, participants gradually responded more slowly to targets associated with the delayed disappearance than to immediately disappearing targets. As the slowdown occurs before the delayed feedback, the results show that participants adjusted their actions based on the target's expected temporal features. These findings were also replicated by Höbl et al. [13] in a less controlled online setting, underscoring the robustness of the so-called slowdown effect and its relevance for HCI contexts.

While both Bogon et al. [1] and Höbl et al. [13] established the slowdown effect, their studies were limited to a relatively short interaction period of 8 game rounds (approx. 40 minutes). Therefore, it remains unclear how the effect develops over an extended period of time and whether it persists once the delay is removed. In this paper, we address these questions through two experiments using the prototype by Höbl et al. [13], each conducted with 50 participants. The first experiment examines the dynamics of the slowdown effect over an extended game-play time of 16 game rounds (approx. 80 minutes). The results show that the slowdown effect further increases over time. In the second experiment, we explore whether learned slowdown behavior persists after the delay is removed. Descriptive data suggest that participants respond more slowly to the target previously associated with delay. Therefore, the slowdown effect may persist for some time before diminishing again. Overall, we demonstrate that users' responses slow down, and consequently, they adapt their behavior over time when interacting with delayed systems. This emphasizes the importance of considering the cognitive impact of delays in interactive systems.

2 Method

We tested two research questions in our experiments. First, in Experiment 1, we examined whether the user slowdown effect caused by system delays changes with prolonged exposure. For analysis, we combined two subsequent game rounds into one block. We hypothesized that users take longer to click on delayed targets than on immediate targets (H1) and that the effect size of the slowdown effect increases across blocks (H2). Second, for Experiment 2, we investigated whether the slowdown effect persists after the delay is removed. We expected the effect to increase over time during the initial delay phase (H3) and to persist for at least one block after the delay removal, gradually decreasing in the no-delay phase (H4). The procedures and analysis plans for both experiments were preregistered on AsPredicted (Experiment 1: aspredicted.org/t4pk-72bk.pdf; Experiment 2: aspredicted.org/5v5c-hqsw.pdf). All raw data and analysis scripts are available on OSF (Experiment 1: osf.io/jkg4e; Experiment 2: osf.io/g4bje).

2.1 Procedure

Both experiments were conducted as online studies using the same game-based pointing task by Höbl et al. [13] (see Figure 1 panel A). After providing informed consent, participants were directed to the experimental website. Participants were instructed to move their cursor to the side target and click it as quickly and accurately as possible. Each trial began with the appearance of a middle target, which participants had to click. After a fixed interstimulus interval of 0.4 s, a target appeared either on the left or right side of the screen. Targets on one side disappeared immediately with the click, whereas targets on the other side disappeared after 0.6 s. The side on which the delay happened was fixed within participants and counterbalanced across participants. Each game round consisted of

60 trials, 30 targets that disappear immediately and 30 targets that disappear after a 0.6 s delay. The order of targets was randomized. Two subsequent rounds were combined into one block. In both experiments, participants completed 960 trials. After finishing the game, participants answered a post-game questionnaire assessing subjective experiences, general observations about the task, and demographic information. All logged data were anonymized.

The two experiments differed only in the final four blocks, in which we either removed the delay or maintained it. In Experiment 1, the delay was present throughout all eight blocks, allowing us to examine the development of the slowdown effect under prolonged exposure. In Experiment 2, the delay was present only during the first four blocks and was removed for the last four blocks, such that both targets provided immediate feedback. This two-phase structure enabled an investigation of the persistence and decay of the slowdown effect after delay removal.

2.2 Participants

Two independent samples were collected for the two experiments using identical inclusion and exclusion criteria. Participants were recruited via Prolific and were required to provide informed consent, speak English fluently, and be aged 18–45. We excluded participants who reported not using a mouse, experienced technical issues, or submitted incomplete data. Additional exclusions were applied when the logged data showed excessive error rates, in accordance with our preregistrations. For each excluded participant, a new participant was recruited until 50 valid datasets were obtained per experiment. For Experiment 1 (Growth), the final sample consisted of 50 participants (24 female, 26 male; 47 right-handed, 3 left-handed), aged 20–45 ($M = 29.38$, $SD = 6.84$). For Experiment 2 (Persistence), the 50 participants (36 male, 14 female) were aged 20–45 years ($M = 28.6$, $SD = 7.02$). Most participants self-assessed as right-handed (47 right-handed, 2 left-handed, 1 ambidextrous). All (included and excluded) participants received £12.75 for the 80-minute-long experiments.

3 Results

We collected a total of 960 trials per participant in both experiments. Following our preregistration, we applied identical trial exclusion criteria. Specifically, we excluded the first trial of each round, trials without a successful hit (Experiment 1: 5.01%; Experiment 2: 6.31%), trials with *response times* (RTs) exceeding 4,000 ms (1: 0.25%; 2: 1.95%), and trials with RTs deviating more than three standard deviations from the participant-specific mean (1: 0.74%; 2: 0.68%). Therefore, the final dataset included 92.31% of all trials in Experiment 1, and 89.5% in Experiment 2).

Data was analyzed in R (version 4.5.0) using repeated-measures ANOVAs and paired *t*-tests. Violations of sphericity were corrected using the Greenhouse-Geisser method, and the results are reported with ϵ estimates. Post-hoc *t*-tests were adjusted using the Benjamini-Hochberg correction. The analysis of error rates is provided in the supplementary material and OSF repositories.

3.1 Experiment 1: Growth of the Slowdown Effect

A 8 (BLOCK) $\times$ 2 (TARGET TYPE) repeated measures ANOVA on *RT* revealed a main effect of BLOCK, $F(3.59, 176.11) = 12.6$, $p < .001$, $\eta_p^2 = 0.21$, $\epsilon = 0.513$. Participants' mean *RT*s decreased from 1160 ms ($SD = 526.33$) in Block 1–2 to 1011 ms ($SD = 445.23$) in Block 15–16. We also found a significant main effect of TARGET TYPE, $F(1, 49) = 19.43$, $p < .001$, $\eta_p^2 = 0.28$. Participants responded more slowly to targets with delayed responses (mean *RT* = 1094 ms, $SD = 458.96$) compared to those with immediate responses (mean *RT* = 1033 ms, $SD = 496.34$). Crucially, we found a significant interaction effect of BLOCK $\times$ TARGET TYPE, $F(3.72, 182.43) = 4.89$, $p = .001$, $\eta_p^2 = 0.09$, $\epsilon = 0.532$. At the beginning of the experiment, the mean difference was 23.3 ms ($SD = 80.41$, $t(49) = 2.05$, $p = 0.046$, $d = 0.29$). By the final block, it expanded to 103.8 ms ($SD = 175.96$, $t(49) = 4.17$, $p < .001$, $d = 0.59$). For a complete overview of all blocks, see Table 1 and Figure 1 panel B. Overall, these findings indicate that while participants generally became faster, the slowdown for targets with delayed disappearance became more pronounced.

Table 1: Mean (M) and standard deviations (SD) of response times (RT, in ms) for Experiment 1. Differences (Diff) are calculated as Delay minus No-Delay. Reported *p*-values are Benjamini–Hochberg corrected. Effect sizes are reported as Cohen's *d*.

Block	No Delay		Delay		Statistics		
	M	SD	M	SD	Diff	<i>p</i>	<i>d</i>
1–2	1148	533	1172	525	23.3	.046	0.29
3–4	1093	526	1124	486	30.9	.040	0.31
5–6	1044	502	1096	462	52.2	.003	0.46
7–8	1042	510	1092	463	49.5	.003	0.47
9–10	1018	490	1082	456	63.7	<.001	0.62
11–12	988	488	1072	444	83.9	.001	0.57
13–14	972	462	1052	422	80.4	.002	0.49
15–16	959	461	1063	427	103.8	<.001	0.59

3.2 Experiment 2: Persistence of the Slowdown Effect

We conducted a 2 (EXPERIMENT PHASE) $\times$ 4 (BLOCK) $\times$ 2 (TARGET TYPE) repeated measures ANOVA on *RT*. The results revealed a significant three-way interaction effect between EXPERIMENT PHASE, BLOCK, and TARGET TYPE: $F(3.59, 176.11) = 12.6$, $p < .001$, $\eta_p^2 = 0.21$, $\epsilon = 0.51$. This indicates that the trend in the BLOCK $\times$ TARGET TYPE effects differs significantly between the delay and no-delay phases. As preregistered and to further investigate the significant three-way interaction, we conducted two 4 (BLOCK) $\times$ 2 (TARGET TYPE) repeated-measures ANOVA on *RT* for each EXPERIMENT PHASE.

3.2.1 Delay Phase. We found a significant main effect of BLOCK, $F(2.11, 103.44) = 4.9$, $p = .008$, $\eta_p^2 = 0.09$, $\epsilon = 0.70$.

There was no significant main effect of TARGET TYPE, $F(1, 49) = 2.6$, $p = .116$, $\eta_p^2 = 0.05$. As well as no significant interaction effect of

BLOCK x TARGET TYPE, $F(3, 147) = 2.1$, $p = .098$, $\eta_p^2 = 0.04$. Although the interaction effect was not statistically significant, the descriptive trends suggest a similar tendency as in previous findings. The difference of RTs grows from -5 ms in Block 1-2 (SD = 100.88), gradually to 40 ms (SD = 154.49) in Block 7-8. Participants descriptively responded more slowly to targets with delayed feedback than to targets with immediate feedback.

3.2.2 No-Delay Phase. We found a significant main effect of BLOCK, $F(2.21, 108.29) = 6.2$, $p = .001$, $\eta_p^2 = 0.113$, $\epsilon = 0.737$. Participants responded faster, with RTs decreasing from 1217 ms (SD = 606.7) in Block 1-2 to 1146 ms (SD = 587.52) in Block 7-8. No significant main effect of TARGET TYPE, $F(1,49) = 1.3$, $p = .263$, $\eta_p^2 = 0.026$, was found. There was also no evidence for a significant interaction effect of BLOCK x TARGET TYPE, $F(3, 147) = 2.6$, $p = 0.054$, $\eta_p^2 = 0.051$. Despite the non-significant interaction effect, the descriptive data (see Table 2 and Figure 1 panel C) suggest a tendency consistent with our hypothesized development. The difference in RTs decreases from 35ms (SD = 138.18) in Block 1-2 to -1ms (SD = 154.49) in Block 7-8. After removing the delay, participants appear to re-adapt to a similar RT for both target types.

Table 2: Mean (M) and standard deviations (SD) of response times (RT, in ms) for Experiment 2. Differences (Diff) are calculated as Delay minus No-Delay. Reported p -values are Benjamini–Hochberg corrected. Effect sizes are reported as Cohen’s d .

Phase	Block	No Delay		Delay		Statistics		
		M	SD	M	SD	Diff	<i>p</i>	<i>d</i>
Delay	1–2	1364	656	1359	665	-5.0	.730	-0.05
	3–4	1318	712	1337	692	19.1	.476	0.13
	5–6	1273	675	1311	627	38.9	.154	0.28
	7–8	1266	698	1308	656	42.3	.154	0.26
NoDelay	1–2	1199	628	1234	590	35.0	.160	0.25
	3–4	1175	601	1214	610	39.5	.160	0.29
	5–6	1167	605	1170	565	3.5	.980	0.02
	7–8	1146	610	1146	571	-0.5	.980	0.00

4 Discussion and Conclusion

In this paper, we investigated how repeated exposure to delayed system feedback shapes user behavior over time, focusing on the growth and persistence of the slowdown effect. Prior work has demonstrated that delays in action effects become integrated into users’ action planning, leading to slower responses over time [1, 13]. We extend this line of research by examining whether this effect strengthens with longer interaction and whether it persists after the delay is removed. We conducted two online experiments using a simple point-and-click task. Participants were instructed to click on targets as quickly as possible. Depending on the target location, they disappeared either immediately or after a fixed 600 ms delay.

Experiment 1 examined whether the slowdown effect stabilizes or continues to increase over extended exposure. The task duration was doubled compared to prior work [1, 13], resulting in 16 rounds of gameplay. Results show that participants responded significantly

slower to targets with delayed disappearance compared to those with immediate feedback. The response time difference increased steadily from 23 ms in the initial block to 104 ms in the final block. We can therefore confirm both H1 and H2. Importantly, the slowdown effect showed no sign of stabilization, suggesting that the action-effect associations continue to strengthen over an extended period. This provides further evidence that repeated exposure to delays can integrate temporal features of effects into users’ action planning and slow their actions. This is in line with ideomotor theory [29] and the formation of action-effect associations [6, 18].

The growing slowdown effect and its consequent change in user behavior raise a critical, subsequent question about the persistence and potential decay of these adaptations. Therefore, in Experiment 2, we investigated whether the slowdown effect persists after the delay is removed. The game was divided into two phases. In the first half of the game, one target type disappeared with a delay, whereas in the last half, all targets disappeared immediately. The slowdown effect evolves statistically differently across the two phases. Subsequent analyses for each phase did not provide statistically significant evidence of a slowdown effect in the first phase, and consequently, there was no statistically significant persistence. We can therefore not confirm H3 and H4.

However, descriptive statistics suggest a pattern that is consistent with persistence. Participants responded on average 40 ms slower to targets with delayed disappearance than to those with immediate feedback. Although this difference was not statistically significant, the descriptive trend is compatible with the possibility that a slowdown effect emerged over the first phase. This difference in response time persisted descriptively for two blocks before diminishing in the second phase, in which all targets disappeared immediately. This pattern aligns with associative learning accounts, which suggest that learned associations are not immediately erased but instead decay gradually as new associations are formed [2, 25]. This also provides further insights into the findings of prior work [20, 23], which found behavioral adaptations to latency even after the delay was removed. Our results suggest that such effects may be understood as the persistence of previously formed action-effect associations. The weaker statistical evidence in Experiment 2 is likely due to increased variability in participant behavior. Compared with Experiment 1 and prior work [1, 13], participants showed longer initial response times (approx. 200 ms) and a higher proportion of outliers (failed trials and trials exceeding 4,000 ms), suggesting reduced focus or engagement. Furthermore, Experiment 2 launched shortly after Experiment 1, using the same recruitment platform and time frame. We strictly excluded previous participants to ensure a fresh sample. This subsequent recruitment may have exhausted the pool of highly interested and motivated participants willing to complete an 80-minute study. This increased noise may have reduced the statistical power to detect the persistence of the slowdown effect. Therefore, future work should reinvestigate this experiment in a more controlled environment. It also remains unclear how the effect develops over multiple days and how it scales with different delays. Nevertheless, our findings reinforce the importance of minimizing delays in interactive systems, especially in time-critical contexts. At the same time, we emphasize that the slowdown effect may be intentionally used in situations where slower actions are desired, e.g., as a (game) mechanic to benefit from a speed-accuracy

trade-off. From a research perspective, a possible persistence of the slowdown effect highlights a critical implication for evaluating interactive systems. When comparing different latency systems, action-effect associations can be formed and continue to influence user behavior even after a latency switch. As a result, measured performance may not only reflect the current system state but also the persistent effect from earlier settings [20, 23]. To address this, future evaluations should carefully counterbalance different latency conditions or provide settings to unlearn previously formed associations.

In conclusion, this paper demonstrates that system delays affect user behavior beyond their immediate effects. They become embedded in users' action planning, slowing down their actions over time. These adaptations may persist even after delays are removed, underscoring the importance of further investigation. HCI designers and researchers should consider the long-term cognitive impacts of system delays. Recognizing and understanding delays as both technical and cognitive factors is therefore essential for the design and evaluation of interactive systems.

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